EXPERIMENT-11
16-BIT MULTIPLICATION

AIM: To write an assembly language program to implement 16-bit multiplication on 8086 processor.

ALGORITHM:

1. Load the first data in HL pair
2. Move content of HL pair to stack pointer.
3. Load the second data in the HL pair and move it to DE
4. Make H register as 0H and L register 0H.
5. Add HL pair and stack pointer.
6. Check for carry if carry increment by 1 else move to next step.
7. Then move E to A and perform or operation with accumulation and register D.
8. The value of operation is zero the solve the value else go to step 3.

PROGRAM:

```
MOV AX, [1100H] - load first number
MOV BX, [1102H] - load second number
MUL BX - Multiply AX by BX.
MOV [1200H], AX - Store higher result.
MOV [1202H], DX - Store higher result.
HLT - Halt program execution.
```

Input:

Address	Data
1100	20
1202	3

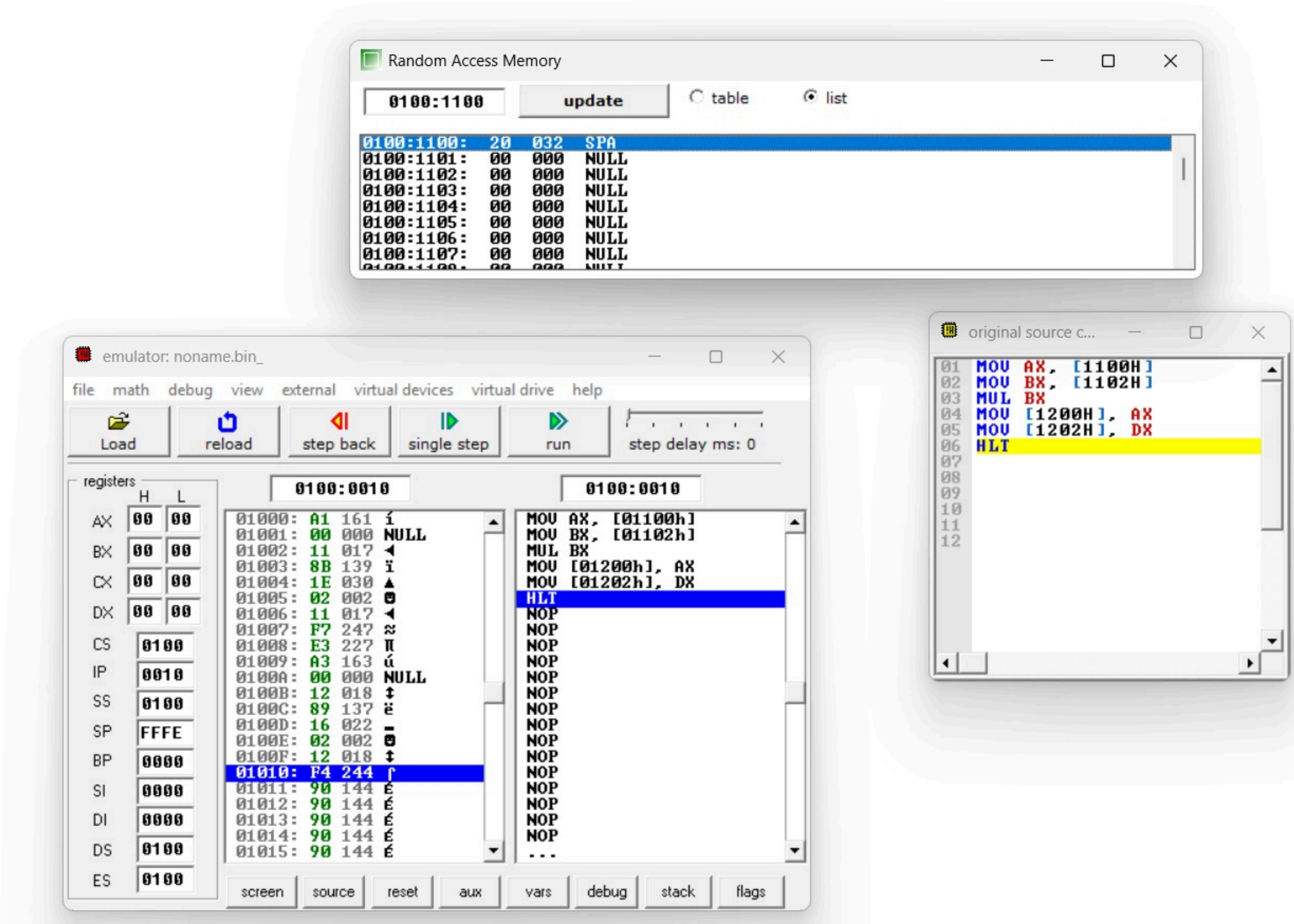
Output:

Address	Data
1200	60
1202	3

0181

Result: Thus the program was executed successfully using 8086 simulator.


MOV [1202H], DX
HLT


Random Access Memory
0100:1100
update
table
list
0100:1100: 20 032 SPA
0100:1101: 00 000 NULL
0100:1102: 00 000 NULL
0100:1103: 00 000 NULL
0100:1104: 00 000 NULL
0100:1105: 00 000 NULL
0100:1106: 00 000 NULL
0100:1107: 00 000 NULL
emulator: noname.bin_
file math debug view external virtual devices virtual drive help
Load
reload
step back
single step
run
step delay ms: 0
registers
H L
AX 00 00
BX 00 00
CX 00 00
DX 00 00
CS 0100
IP 0010
SS 0100
SP FFFE
BP 0000
SI 0000
DI 0000
DS 0100
ES 0100
0100:0010
0100:0010
01000: A1 161 í
01001: 00 000 NULL
01002: 11 017
01003: 8B 139 ï
01004: 1E 030
01005: 02 002
01006: 11 017
01007: F7 247 ≈
01008: E3 227 π
01009: A3 163 ú
0100A: 00 000 NULL
0100B: 12 018
0100C: 89 137 ë
0100D: 16 022
0100E: 02 002
0100F: 12 018
01010: F4 244 ⌠
01011: 90 144 É
01012: 90 144 É
01013: 90 144 É
01014: 90 144 É
01015: 90 144 É
MOV AX, [01100h]
MOV BX, [01102h]
MUL BX
MOV [01200h], AX
MOV [01202h], DX
HLT
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
...
screen
source
reset
aux
vars
debug
stack
flags
original source c...
01 MOV AX, [1100H]
02 MOV BX, [1102H]
03 MUL BX
04 MOV [1200H], AX
05 MOV [1202H], DX
06 HLT
07
08
09
10
11
12